THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 7: **Measurement of Time and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A train 150 m long crosses a platform 250 m long in 20 s. The speed of the train is: 1

(a) 12.5 m/s **(b)** 7.5 m/s **(c)** 8 m/s **(d)** 20 m/s

2. Two cyclists A and B start from the same point. A goes at 15 km/h and B at 20 km/h. After 3 h, B is ahead of A by: 1

(a) 60 km **(b)** 15 km **(c)** 5 km **(d)** 35 km

3. A pendulum was timed for 10 oscillations: 18.2 s, 18.0 s, 18.4 s. The best estimate of T is: 1

(a) 0.55 s **(b)** 1.82 s **(c)** 18.2 s **(d)** 1.8 s

4. Two trains 100 m and 200 m long move in opposite directions at 20 m/s and 10 m/s. They cross in: 1

(a) 15 s **(b)** 30 s **(c)** 5 s **(d)** 10 s

5. A train leaves at 6 a.m. at 60 km/h. A second train leaves at 8 a.m. at 90 km/h on the same track. It catches the first at: 1

(a) 2 p.m. **(b)** 8 a.m. **(c)** 10 a.m. **(d)** 12 noon

6. A sprinter's reaction time is 0.2 s and a stopwatch is started after seeing the start. The measured 10 s may be: 1

(a) double **(b)** more than true

(c) exact **(d)** less than true by about 0.2 s

7. A man runs round a circular track of length 400 m in 100 s. His average speed is: 1

(a) 400 m/s **(b)** 0 m/s **(c)** 40 m/s **(d)** 4 m/s

8. A man goes from A to B at 4 km/h and returns at 6 km/h. The time taken in the round trip is 5 h. The distance AB is: 1

(a) 24 km **(b)** 5 km **(c)** 12 km **(d)** 6 km

9. A student measures one oscillation of a pendulum with a stopwatch of reading to 0.1 s and gets 2.0 s. The error is about: 1

(a) 50 per cent **(b)** up to 5 per cent

(c) up to 0.5 per cent **(d)** nil

10. His displacement after one full round is: 1

(a) 400 m **(b)** 4 m **(c)** 0 **(d)** 100 m

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Calculate the speed of a train in km/h if it crosses a 300 m bridge in 30 s and its length is 200 m. 2

12. Why can't the average speed of a round trip be the average of two speeds? 2

13. Why is acceleration not required to solve problems on uniform speed? 2

14. A student recorded: 10 oscillations in 15.2 s, 15.0 s, 15.4 s. Find the mean time period. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A train 120 m long crosses a pole in 12 s and a platform in 30 s. Find its speed and the length of the platform. 3

16. At 5 a.m. a bus leaves a town at 40 km/h. A car leaves the same town on the same road at 6 a.m. at 60 km/h. When does the car catch up? 3

17. The speed of sound is 340 m/s. How long does the sound take to travel 1.7 km? 3

OR

A person goes from A to B at 40 km/h and returns at 60 km/h. Find the average speed. 3

18. Explain the use of the 'seconds pendulum' in old clocks. 3

OR

A car's speed is 90 km/h. How far does it go in 1 s in metres? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student plots the following data: time (s): 0, 5, 10, 15, 20; distance (m): 0, 10, 20, 20, 50. Describe the stages of motion, find the speed in each stage and the average speed. 5

OR

A car's odometer reads 25 340 km at 9:00 a.m. and 25 420 km at 11:00 a.m. Its speedometer reads 60 km/h at 9:30 a.m. and 20 km/h at 10:30. Explain the difference between average and instantaneous speeds. 5

20. Design an experiment to show how the time period of a pendulum varies with its length. Specify measurements, tables and expected relation. 5

OR

A race track is 400 m. A runner completes 5 laps in 10 minutes. Find the speed in m/s and km/h and the displacement. 5

END OF QUESTION PAPER — Total: 40 Marks